

Hacettepe University - Computer Engineering CMP720 Embedded System Design - Spring 2026

Final Project Proposal

Group Members (Name Surname + Student ID)

- Oğulcan Uğuroğlu — 24120210
- Barış Büyükyılmaz — 24120211

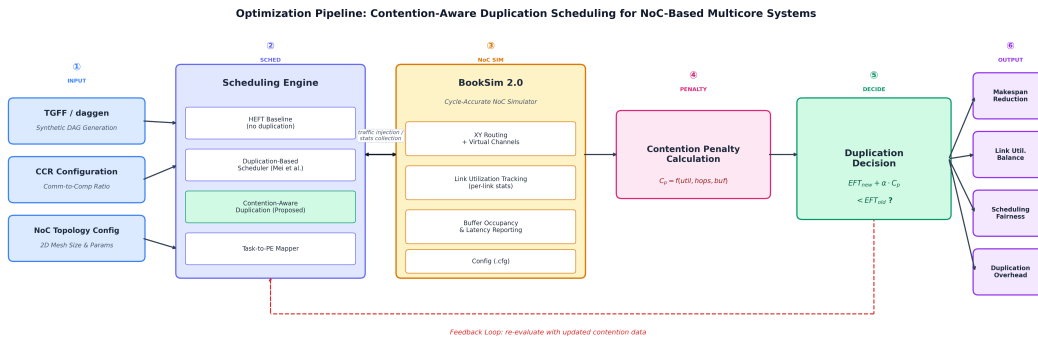
Project Topic Title

Contention-Aware Duplication Scheduling for Network-on-Chip Based Multicore Systems

Project Track (choose one)

- ☐ Implementation Track
- ☒ Optimization Track

Network-on-Chip (NoC) based multiprocessor systems-on-chip (MPSoCs) are widely used in modern embedded and high-performance computing systems due to their scalability and modular communication infrastructure. Efficient task scheduling in such systems is critical for minimizing execution time while maintaining balanced utilization of computational and communication resources. Duplication-based scheduling is a commonly used technique that replicates tasks across processors to reduce communication delays between dependent tasks. However, most existing approaches primarily focus on computation and communication latency while neglecting the impact of network contention. As NoC topologies scale to larger sizes (e.g., 8×8 mesh networks or beyond), link contention and communication congestion may significantly degrade overall system performance. This project proposes a contention-aware duplication scheduling approach that integrates network-level congestion metrics into duplication decisions. In classical duplication-based scheduling, a task is duplicated if the new earliest finish time (EFT) is smaller than the original. In this work, the duplication decision is augmented with a contention penalty term reflecting network congestion characteristics such as link utilization, hop count, and buffer pressure. This penalty prevents excessive duplication in congested network regions and improves overall communication efficiency. The system leverages existing tools alongside custom-developed components. Synthetic directed acyclic graphs (DAGs) are generated using established task graph generators such as TGFF or daggen with varying communication-to-computation ratios (CCR). Network behavior is evaluated using BookSim 2.0, a cycle-accurate NoC simulator that provides realistic link utilization, buffer occupancy, and latency statistics for 2D mesh topologies. The scheduling engine implements three strategies for comparison: a baseline HEFT algorithm without duplication, a classical duplication-based scheduler following the approach of Mei et al., and the proposed contention-aware heuristic. Performance is evaluated using metrics including application makespan, link utilization balance, scheduling fairness, and duplication overhead. The project proceeds in three stages: first, integrating the task graph generator and BookSim into a baseline scheduling framework; second, developing and incorporating the contention-aware duplication heuristic; and third, conducting experimental evaluation across different CCR values and NoC sizes. This study aims to provide insights into improving scheduling strategies for NoC-based multicore systems by incorporating congestion awareness into task duplication decisions.



Modern embedded systems are increasingly deployed in safety-critical and performance-sensitive domains such as autonomous vehicles, medical imaging devices, industrial automation, and telecommunications infrastructure. These systems rely on NoC-based multicore processors to execute complex workloads under strict timing and resource constraints. As the number of processing elements in such systems continues to grow, the efficiency of task scheduling becomes a decisive factor in meeting real-time deadlines, reducing energy consumption, and ensuring system reliability. Inefficient scheduling decisions can lead to excessive network congestion, resulting in increased communication latency, higher power consumption, and potential violations of timing guarantees. In safety-critical applications such as Advanced Driver Assistance Systems (ADAS) or real-time medical signal processing, missed deadlines caused by network contention can have serious consequences for human safety. Similarly, in data center processors and edge computing devices, poor scheduling contributes to wasted energy and reduced throughput, both of which carry direct environmental and economic implications. Task duplication is a powerful technique for reducing inter-processor communication overhead, but each unnecessary duplication consumes additional processing and communication resources. When duplication decisions are made without considering network congestion, the resulting increase in traffic can worsen contention and negate the intended performance benefit. This problem becomes particularly severe as NoC topologies scale beyond 4×4 configurations. The proposed approach benefits embedded system designers through more realistic scheduling models, and application developers through more predictable execution times. From a broader perspective, improved scheduling efficiency translates to lower energy consumption per computation, contributing to the sustainability of computing infrastructure. Impact can be measured through makespan reduction percentages, decrease in unnecessary duplications, link utilization balance improvements, and energy overhead analysis across different NoC sizes and workload configurations.

Duplication-based scheduling has been extensively studied in the context of heterogeneous computing systems. Mei et al. [1] proposed a resource-aware scheduling algorithm that reduces unnecessary task duplications by evaluating whether duplication yields a meaningful reduction in execution time. Their approach uses a greedy duplication strategy that considers computation and communication costs but does not explicitly model network-level contention effects. Tang et al. [2] addressed the problem of optimizing duplication-based schedules specifically for NoC-based MPSoCs. They formulated the scheduling problem as an Integer Linear Programming (ILP) model that accounts for communication contention on network links. However, in their formulation, the duplication decisions are treated as fixed inputs to the optimization, and the contention model is applied only during the ordering and timing phase rather than during the duplication decision itself. On the dynamic scheduling side, Paul [3] proposed an adaptive strategy for resource allocation in multitasking NoC-based multicore systems. This work introduces a hybrid approach that combines static mapping with dynamic adjustment based on runtime workload characteristics, providing flexibility for systems with varying task arrival patterns. For heterogeneous task mapping, Huang et al. [4] developed a scalable ILP-based approach that jointly considers load balance and communication optimization on MPSoC platforms. Their method demonstrates effective scaling to larger problem instances through decomposition techniques, though it focuses on mapping rather than duplication decisions. The proposed project builds upon these foundations by integrating network contention metrics directly into the duplication decision process, an aspect that has not been jointly addressed in the existing literature.

References

- [1] J. Mei, K. Li, A. Ouyang, and K. Li, “A resource-aware scheduling algorithm with reduced task duplication on heterogeneous computing systems,” *The Journal of Supercomputing*, vol. 68, no. 3, pp. 1347–1377, 2014.
- [2] X. Tang, X. Liao, J. Jia, and K. Li, “Optimization of duplication-based schedules on network-on-chip based multi-processor system-on-chips,” *IEEE Transactions on Parallel and Distributed Systems*, vol. 26, no. 4, pp. 1171–1187, 2015.
- [3] S. K. Paul, “An adaptive strategy for dynamic resource allocation and scheduling for multi-tasking NoC based multicore systems,” in *Proceedings of the 28th International Symposium on VLSI Design and Test (VDATE)*, Springer, 2024.
- [4] P.-Y. Huang, Y.-W. Chen, and T.-H. S. Liang, “A scalable and adaptable ILP-based approach for task mapping on MPSoC considering load balance and communication optimization,” *IEEE Access*, vol. 11, pp. 1–15, 2023.